

Super Simple INTERCOM

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This intercom circuit based on IC LM380 audio amplifier requires very few external components. So the circuit is very easy to assemble.

Circuit and working

Circuit diagram of the intercom is shown in Fig. 1. In addition to audio amplifier LM380 (IC1), it uses a condenser microphone (MIC1), an 8-ohm, 0.5W speaker and a few other components.

PARTS LIST

Semiconductors:

IC1 - LM380 audio amplifier

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1 - 10-kilo-ohm
R2, R3 - 4.7-kilo-ohm
VR1 - 100-ohm potmeter

Capacitors:

C1 - 0.1 μ F ceramic disk
C2 - 0.01 μ F ceramic disk
C3 - 10 μ F, 25V electrolytic
C4 - 100 μ F, 25V electrolytic

Miscellaneous:

MIC1 - Condenser microphone
CON1 - 2-pin terminal connector
LS1 - 8-ohm, 0.5W speaker
S1 - On/off switch
S2 - Push-to-on tactile switch
- 9V regulated power supply or 9V battery with connector

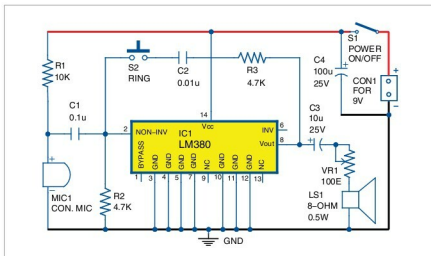


Fig. 1: Circuit diagram of super-simple intercom

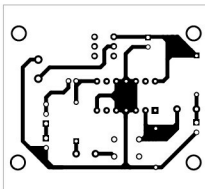


Fig. 2: Actual-size PCB layout for the intercom

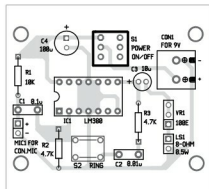


Fig. 3: Components layout for the PCB

Assemble the same circuit on two separate units. In order to use these units as intercom, extend the output (LS1) of the first unit to the second unit placed in a remote location and vice versa. Set the required sound level by adjusting potmeter VR1. Press switch S2 momentarily to generate audio tone in the speaker (LS1).

The circuit works off a 9V DC battery. Alternatively, you can also use 9V unregulated power supply.

Construction and testing

An actual-size PCB layout for the intercom is shown in Fig. 2 and its components layout in Fig. 3.

To test the circuit, you need two units of the same circuit. Extend LS1 from the first unit to the second unit, and LS2 (not shown here) from the second unit to the first unit.

To talk over the intercom, press switch S2 to ring up the person sitting at the remote side where the

second unit is placed. Then you can speak into microphone MIC1. The person at the other end should be able to hear you through LS1. Similarly, when he rings you up, you should be able to hear him through LS2.

Use shielded wires for connections to both the units. **EFY**



Pradeep G. is a regular contributor to EFY